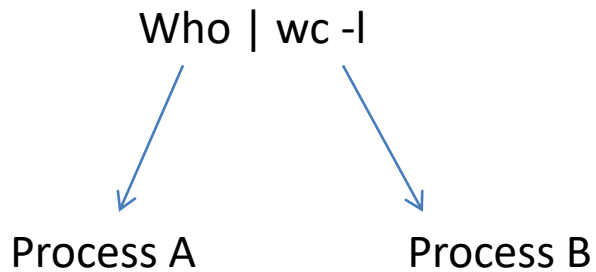


Pipe

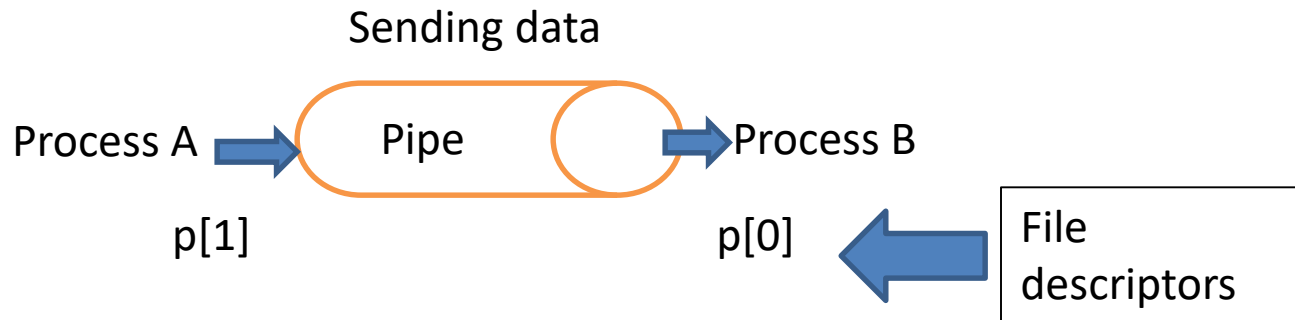
Pipe

- Interprocess communication primitive



Pipe

- Interprocess communication primitive



```
Main()
{
    int p[2];
    pipe(p);
    printf("%d %d",p[0],p[1]);
}
```

Pipe returns -1, if unsuccessful

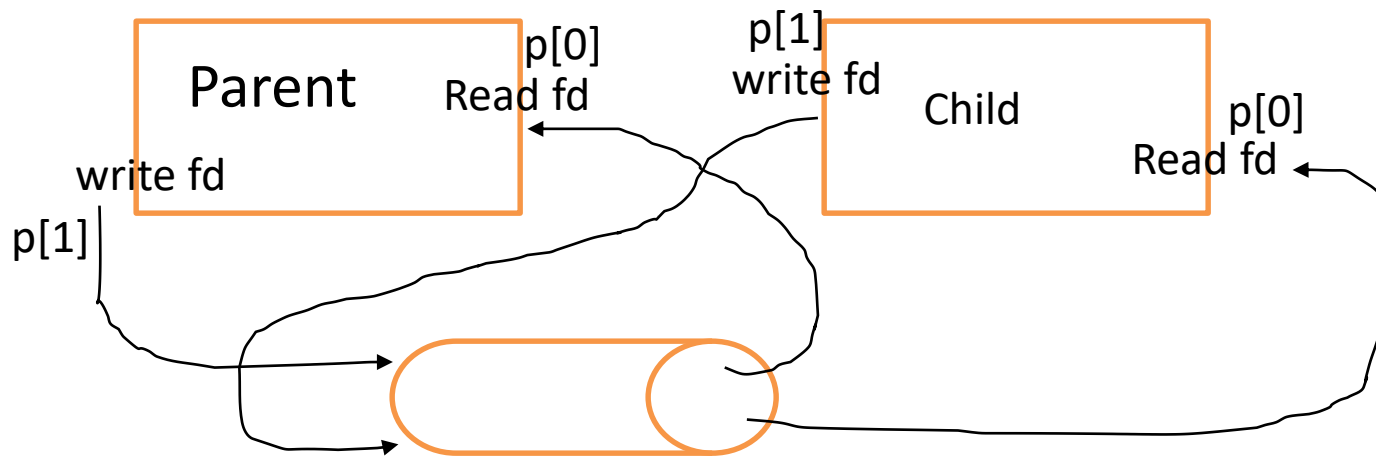
File descriptor table

File descriptor (integer)	File name
0	stdin
1	stdout
2	stderr

Use `open()`, `read()`, `write()` system calls to access files

`Open()` creates a file and returns `fd` (minimum value)

```
fd=open(path, O_WRONLY|O_CREAT)
```



```
Main()
{
```

```
    char *msg="hello";
    char buff[MAX];
    pipe(p);
    pid=fork();
```

```
#define MAX ****
Main()
{
    char *msg="hello";
    char buff[MAX];
    pipe(p);
    pid=fork();
    if(pid>0)
        write(p[1], msg1, MAX);
    else
    {
        sleep(1);
        read(p[0],buf, MAX);
        printf("%s", buff);
    }
}
```

Anybody can write (parent or child)

```
#define MAX ****
main()
{
    char *msg="hello";
    char buff[MAX];
    pipe(p);
    pid=fork();
    if(pid>0)
        write(p[1], msg1, MAX);
    else
    {
        sleep(1);
        write(p[1], msg1, MAX);
        for(i=0;i<2;i++)
        {
            read(p[0],buf, MAX);
            printf("%s", buff);
        }
    }
}
```

```
main()
{
    char *msg="hello";
    char buff[MAX];
    pipe(p);
    pid=fork();
    if(pid==0)
        printf("child exiting")
    else
    {
        read(p[0],buf, MAX);

    }
}
```

Read will wait
since write end of
parent is open.

Role of close()

```
main()
{
    char *msg="hello";
    char buff[MAX];
    pipe(p);
    pid=fork();
    if(pid==0)
    {
        printf("child exiting")
    }
    else
    {
        close(p[1]);
        read(p[0],buf, MAX);
    }
}
```

- Closing write end
- Read will return immediately.

Role of close()

```
main()
{
    char *msg="hello";
    char buff[MAX];
    pipe(p);
    pid=fork();
    if(pid==0)
    {
        sleep(5);
        printf("child exiting")
    }
    else
    {
        close(p[1]);
        read(p[0],buf, MAX);
    }
}
```

What will happen?

Closing the read end

```
main()  
{
```

```
    char *msg="hello";
```

```
    char buff[MAX];
```

```
    pipe(p);
```

```
    pid=fork();
```

```
    if(pid==0)
```

```
    {
```

```
        printf("child exiting")
```

```
    }
```

```
    else
```

```
    {
```

```
        write(p[1],buf, MAX);
```

```
    }
```

```
}
```

Write will return
successfully

```

main()
{
    char *msg="hello";
    char buff[MAX];
    pipe(p);
    pid=fork();
    if(pid==0)
    {
        printf("child exiting")
    }
    else
    {
        sleep(1);
        close(p[0]);
        write(p[1],buf, MAX);
    }
}

```

- All the read ends are closed!
- Write returns -1
- Kernel generates SIGPIPE signal
- Terminates with "Broken pipe" message

int dup(int *oldfd*);

The dup(fd) system call copies the file descriptor entry of fd into the FIRST EMPTY ENTRY in the file descriptor table.

File descriptor (integer)	File name
0	stdin
1	stdout
2	stderr

```
int main()
{
    // open() returns a file descriptor file_desc to a
    // the file "dup.txt" here"

    int file_desc = open("dup.txt", O_WRONLY | O_APPEND);

    if(file_desc < 0)
        printf("Error opening the file\n");

    // dup() will create the copy of file_desc as the copy_desc
    // then both can be used interchangeably.

    int copy_desc = dup(file_desc);

    // write() will write the given string into the file
    // referred by the file descriptors

    write(copy_desc, "This will be output to the file named dup.txt\n", 46);

    write(file_desc, "This will also be output to the file named dup.txt\n", 51);

    return 0;
}
```

File descriptor table

File descriptor (integer)	File name
0	stdin
1	stdout
2	stderr

Use open(), read(), write() system calls to access files

Open() creates a file and returns fd (minimum value)

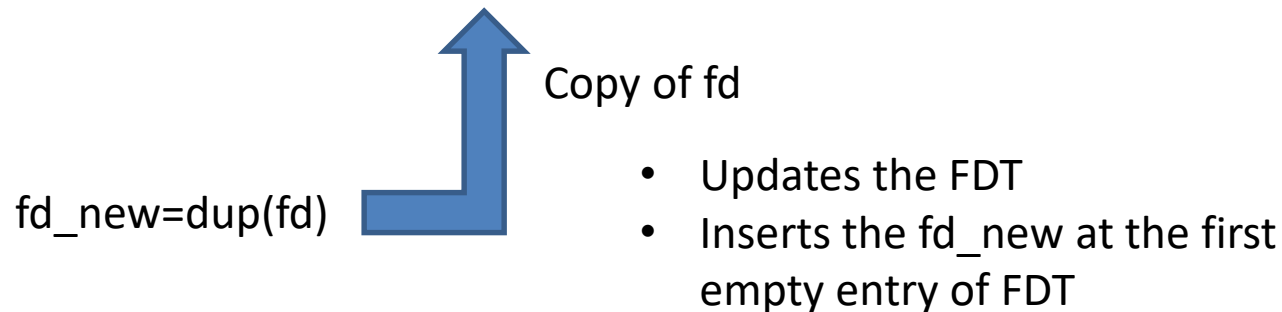
```
fd=open(path, O_WRONLY|O_CREAT)
```

Think what happens in case of redirection?

```
ls>file
```

File descriptor table

File descriptor (integer)	File name
0	stdin
1	stdout
2	stderr



int dup(int fd)


```
#include<stdio.h>
#include<fcntl.h>
int main()
{
    int old,new;
    old=open("input.txt", O_WRONLY|O_CREAT);

    //    new=dup2(old,1);

    close(1);
    new=dup(old);
    printf("hello world\n");
}
```

```
#include<fcntl.h>
int main()
{

    int fd[2],i,f,g,x, pid;
    char buf[20];
    pipe(fd);

    pid=fork();
    if(pid==0)
    {
        close(1);
        dup(fd[1]);
        printf("hello\n");
    }
    else
    {
        read(fd[0],buf,10);
        printf("%s\n",buf);
    }

}
```

popen()

- popen()
- FILE *popen(char *command, char *type)

```
FILE *fp;  
int status;  
char path[PATH_MAX];  
fp = popen("ls -l", "r");  
if (fp == NULL) /* Handle error */;  
  
while (fgets(path, PATH_MAX, fp) != NULL)  
    printf("%s", path);  
status = pclose(fp);
```

Named pipe or FIFO

- Named pipe or FIFO
- /etc/mknod testpipe p

```
char * phrase = "Stuff this in your pipe  
and smoke it";
```

```
int main ()
```

```
{ int fd1;  
fd1 = open ( "mypipe", O_WRONLY );  
write (fd1, phrase, strlen ( phrase)+1 );  
close (fd1);
```

```
}
```

```
int main ()
```

```
{  
int fd1;
```

```
char buf [100];
```

```
fd1 = open ( "mypipe", O_RDONLY );  
read ( fd1, buf, 100 );  
printf ( "%s\n", buf );  
close (fd1);
```

```
}
```

- **int mknod(const char **pathname*, mode_t *mode*, dev_t *dev*);**
- **mknod("/tmp/MYFIFO", S_IFIFO|0666, 0)**